ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MULTIMEDIA BASICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Multimedia Basics paper 1 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Accept answers in the student's own words if the meaning is correct, and other correct examples. Use consistent accuracy in calculations.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 C ✓ 1.1.3 A ✓ 1.1.4 B ✓ 1.1.5 D ✓ 1.1.6 A ✓ 1.1.7 C ✓ 1.1.8 C ✓ 1.1.9 A ✓ 1.1.10 D ✓ (10)

1.2 1.2.1 False ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 E ✓ 1.3.2 D ✓ 1.3.3 G ✓ 1.3.4 C ✓ 1.3.5 H ✓ 1.3.6 F ✓ 1.3.7 B ✓ 1.3.8 I ✓ 1.3.9 A ✓ 1.3.10 J ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Text ✓; images / graphics ✓; audio / sound ✓; video ✓; animation ✓ (5)

2.2 Any FOUR areas with examples (1 + 1 each): education ✓ – e-learning videos, interactive lessons ✓; entertainment ✓ – games, films, music videos ✓; advertising / marketing ✓ – social media adverts, digital billboards ✓; business ✓ – presentations, training ✓; medicine – 3D models for training; tourism – virtual tours; news websites (8)

2.3 Linear multimedia plays from start to end without the user controlling the order ✓, e.g. a film / video advert ✓; non-linear (interactive) multimedia lets the user choose what to view and in which order ✓, e.g. a website, game or kiosk ✓. (4)

2.4 Any THREE: makes learning more interesting / keeps attention ✓; caters for different learning styles (seeing, hearing, doing) ✓; difficult concepts can be shown visually / simulated ✓; learners can work at their own pace; can be repeated (3)

[20]

QUESTION 3

3.1 (a) Raster: made of pixels ✓; vector: made of mathematical paths / lines, curves and shapes ✓ (b) Raster becomes blurry / pixelated when enlarged ✓; vector keeps its quality at any size ✓ (c) Raster: photographs ✓; vector: logos, icons, illustrations ✓ (6)

3.2 $1\,200 \times 800 = 960\,000$ pixels ✓; $\times$ 24 bits $\div$ 8 = 3 bytes per pixel $\rightarrow$ 2 880 000 bytes ✓; $\div (1\,024 \times 1\,024)$ ✓ = 2,75 MB ✓ (4)

3.3 (a) JPEG ✓ – good compression for photographs with many colours, small file loads fast ✓ (b) PNG ✓ – supports transparency and keeps sharp edges (lossless) ✓ (accept SVG for a vector logo) (4)

3.4 Printers mix cyan, magenta, yellow and black inks ✓; RGB colours (light) may print differently, so CMYK gives a more accurate result ✓. (2)

3.5 Resolution is the number of dots (pixels) per inch ✓ – how much detail an image has ✓. Paper is viewed close up and printers need many dots per inch to look sharp ✓, while screens display fewer pixels per inch (about 72–150 ppi) ✓. (4)

[20]

QUESTION 4

4.1 Uncompressed video files are very large ✓; compression saves storage space and allows faster uploading / streaming ✓. (2)

4.2 Any THREE: video camera / smartphone ✓; microphone ✓; computer with editing software ✓; tripod; lights; headphones; external hard drive (3)

4.3 Any TWO: sample rate ✓; bit depth ✓; quality of the microphone; background noise / recording environment; compression / bit rate (2)

4.4 It plans each scene in sketches before filming ✓, showing camera angles, actions and dialogue ✓, which saves time and money and helps the team to agree on the final product ✓. (3)

[10]

QUESTION 5

5.1 Any FOUR principles, named ✓ and explained ✓: contrast – important text stands out (e.g. dark text on light background); alignment – elements are lined up neatly; repetition / consistency – same fonts and colours throughout; proximity – related items grouped together; balance – weight spread evenly; hierarchy / emphasis – the most important information (shop name, special) is biggest; white space – avoids clutter (8)

5.2 Copyright is the legal right of the creator to control how his or her work is copied and used ✓✓. Any TWO legal ways: create your own images ✓; use royalty-free / Creative Commons images with attribution ✓; buy a licence from a stock-photo site; get written permission (4)

5.3 Any FOUR in a logical order ✓✓✓✓: identify the client's needs / purpose and target audience; plan the content (script, storyboard, flowchart); gather / create the media; develop / build the product; test and get feedback; deliver / publish; evaluate (4)

5.4 Any TWO: graphic designer ✓; video editor ✓; animator; web designer; sound engineer; photographer; game developer; UX designer (2)

5.5 Any TWO: add subtitles / captions to videos ✓; add alt text to images ✓; use clear, large fonts and good colour contrast; allow keyboard navigation; provide audio descriptions (2)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE